

ELECTROPHYSIOLOGICAL CHANGES INDUCED BY PARADOXICAL SLEEP DEPRIVATION AND LITHIUM CHLORIDE POISONING IN RATS

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SUMMARY

In an attempt to analyze the disruption of conditioned taste aversion (CTA) caused by pre-acquisition paradoxical sleep deprivation (PSD), the effect of poisoning (0.15 M LiCl, 4% body weight) on the sleep–wakefulness pattern was studied in 12 rats with chronically implanted electrodes. The polygraphic recording showed that poisoning reduced the total sleep time in the subsequent 3 h from 57 to 42% and REM sleep from 6 to 2%. The lithium chloride effect was still more pronounced after 24-h REM sleep deprivation: the total sleep time decreased from 67 to 46% and REM sleep from 13 to 2%. The change of sleep–wakefulness pattern, most pronounced during the first hour of poisoning, gradually returned to normal but REM sleep was significantly reduced even 3 h after poisoning. It is concluded that 24-h PSD does not alleviate the subsequent poisoning. On the other hand, the lithium chloride induced reduction of post-acquisition REM sleep may enhance the learning impairment caused by pre-acquisition PSD.

INTRODUCTION

Assumptions about the behavioral role of paradoxical sleep (PS) are based on two lines of evidence^{4,6}: learning is impaired by pre-acquisition (and sometimes also post-acquisition) paradoxical sleep deprivation (PSD) and incidence of PS is increased in the post-acquisition period. Although there are many exceptions to the above statements, most experiments support the hypothesis⁹ that PS provides the brain with mediators and/or other substances, an adequate supply of which is an indispensable prerequisite of memory processes. Danguir and Nicolaidis² have recently shown that pre-acquisition but not post-acquisition PSD impairs learning of the conditioned taste

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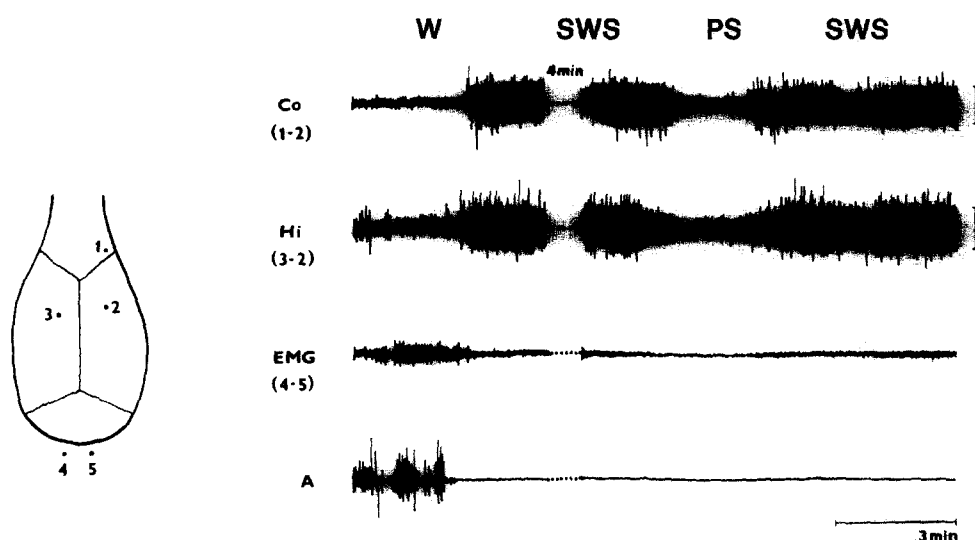


Fig. 1. Example of the polygraphic recording of the sleep-wakefulness pattern in a freely moving rat. The position of the electrodes is indicated in the brain scheme. Co, electrocorticogram; Hi, electrohippocampogram; EMG, electromyogram of neck muscles; A, actogram. The letters W, SWS and PS indicate the sections of the record identified as wakefulness, slow wave sleep and paradoxical sleep, respectively. Calibration 200 μ V.

aversion (CTA) in rats. This finding was confirmed and extended by Venkatakrishna-Bhatt et al.¹¹, who suggested that PSD impairs the initial processing of the gustatory stimulus and formation of the short-term gustatory trace rather than the formation of the longterm CTA engram.

Unlike other classical and instrumental conditioning paradigms, CTA learning employs a US (lithium chloride poisoning) which may induce long-lasting arousal and cause prolonged modification of the sleep-wakefulness pattern. Unless these effects of poisoning are known, the interpretation of the PSD influence on CTA acquisition remains speculative. The aim of the present study was to assess the changes of the electrophysiologically monitored sleep-wakefulness pattern caused by the standard lithium chloride (LiCl) poisoning employed in CTA experiments under control conditions and after 24-h PSD.

METHODS

Animals

The experiments were performed in 12 male hooded rats aged 3 months. The animals were housed in group cages (5 rats/cage) where they had free access to food and water.

Surgery

The rat was anaesthetized with Nembutal (40 mg/kg) and trephine openings (1 mm in diameter) were made above the frontoparietal cortex of the right hemisphere (AP - 2, R2; AP + 2, R2) and above the left dorsal hippocampus (AP 2.5, L1.5) (Fig. 1). The animal was then fixed in the head holder on the stereotaxic apparatus. A pair of

twisted nichrome wires (70 μm in diameter) was inserted along the occipital bone into the exposed neck muscles. Ball tipped silver wires (200 μm in diameter) were placed upon the exposed dura and a varnished stainless steel wire (200 μm) was introduced into the dorsal hippocampus (3.5 mm below skull surface). The electrodes were fixed to the skull with acrylate and anchoring screws and connected to a miniature 5-pin transistor socket, also embedded in acrylate. The experiment started 3–5 days after surgery and continued for 2 weeks.

Procedure

After the rats were habituated to handling and to wearing the flexible cable attached to the head mounted socket, they were put into a circular glass beaker (17.5 cm diameter, 24.5 cm high) placed in a dimly lit, electrically screened, sound attenuating box (70 $\times$ 50 $\times$ 60 cm) and connected to the input of a 4-channel polygraph. Three channels were employed for recording the corticogram, hippocampogram and neck muscle electromyogram with the band pass of the respective amplifiers limited by appropriate setting of filters. The fourth channel, the input of which was connected to a free wire, loosely wrapped around the input cable recorded the movement of the rat. The pen-writer of the polygraph moved the paper with the transport rate of 15 mm/min. Example of the recording is shown in Fig. 1.

Recording started 10–30 min after placing the animal into the recording chamber and continued for at least 4 h (from 1 p.m. to 5 p.m.). The first recording was always made under control conditions (a). After an interval of 2–4 days recording was repeated under one of three experimental conditions. (b) Lithium chloride poisoning: After the first hour of recording the rat was restrained by hand, injected with 0.15 *M* LiCl (4% body weight) and returned into the recording box. (c) Paradoxical sleep deprivation: Recording was preceded by a 24-h stay of the animal on a circular pedestal (7 cm diameter, 7 cm high) surrounded by electrified grid floor. Before being placed into the recording chamber, the animals were allowed 30-min access to food and water. (d) Paradoxical sleep deprivation and LiCl poisoning: Sleep deprived animals (c) were injected with LiCl (b).

The above experimental conditions were applied in random order, with sufficient intervals (3–4 days) providing for full recovery from the LiCl poisoning or PSD after-effects.

Evaluation of the polygraphic records

The total recording time was split into three classes according to the following criteria (Fig. 1).

Wakefulness. Desynchronized, low voltage electrocorticogram, theta rhythm or low irregular activity in the hippocampogram, high neck muscle electromyogram, movements in the actogram.

Slow wave sleep. Irregular high amplitude waves in the corticogram and hippocampogram, absence of movements in the actogram but signs of muscle activity in the electromyogram.

Rapid eye movement (REM) (paradoxical) sleep. Desynchronized low voltage

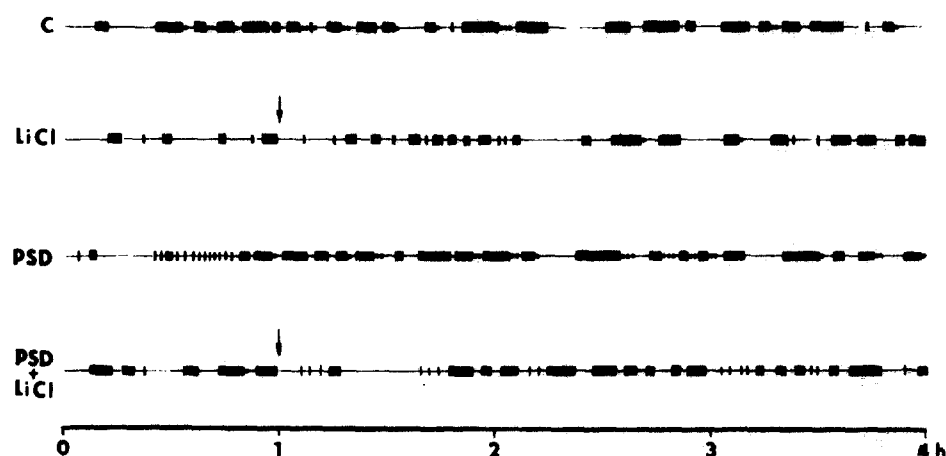


Fig. 2. Typical sleep patterns observed in the same rat under control conditions (C), during lithium chloride poisoning (LiCl), after 24-h paradoxical sleep deprivation (PSD), and under the combined effect of paradoxical sleep deprivation and poisoning (PSD + LiCl). The arrow indicates lithium chloride injection. Time (in hours) since the onset of recording. Symbols: Thin line, wakefulness; thick bar, slow wave sleep and dots, paradoxical sleep.

electrocorticogram, regular theta rhythm in the hippocampogram, absence of movements in the actogram and maximum reduction of electromyogram indicating relaxation of neck muscles.

The percentage of the total recording time spent in wakefulness, slow wave sleep and paradoxical sleep was established at 1-h intervals throughout the recording. The PS duration was also expressed as percentage of the total sleep time (slow wave sleep plus paradoxical sleep).

RESULTS

The small dimension of the beaker restricted movements of the animal who remained relatively quiet throughout the recording session. A typical experiment is shown in Fig. 2 which also illustrates the notation of the various components of the sleep-wakefulness pattern. All results are summarized in Figs. 3 and 4 and based on 6 rats which yielded satisfactory recordings during all stages of the experiment. A $2 \times 2 \times 6$ factorial design was used in the statistical analysis of results.

(a) Control conditions

Wakefulness was predominant during the first hour of recording whereas sleep and particularly REM sleep was low. The sleep-wakefulness pattern stabilized over the subsequent 3 h.

(b) Lithium chloride poisoning

Intraperitoneal injection of LiCl caused prolonged arousal which increased the incidence of wakefulness in the first hour after poisoning. Waking time gradually

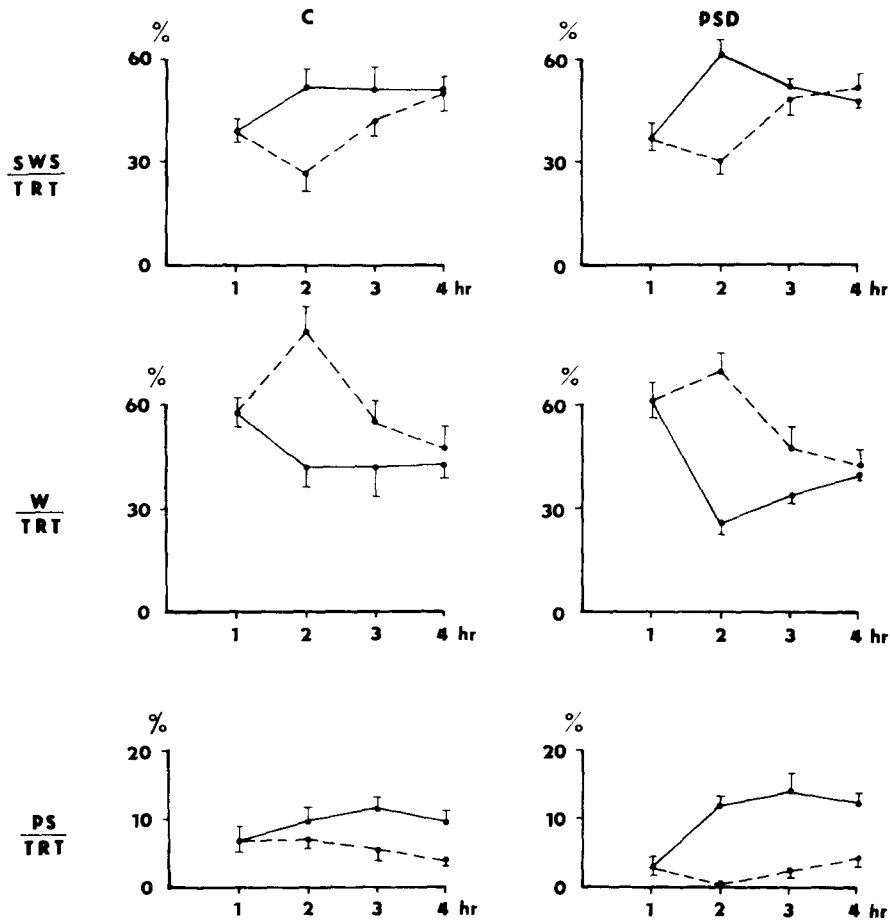


Fig. 3. The effect of paradoxical sleep deprivation and lithium chloride poisoning on the sleep-wakefulness pattern in rats ($n = 6$). The ordinates represent percentage of the total recording time (TRT) covered during the preceding hour by slow wave sleep (SWS), wakefulness (W) or paradoxical sleep (PS) under control conditions (C) and after 24-h paradoxical sleep deprivation (PSD). Full line, untreated rats. Dashed line, rats injected with LiCl (0.15 M, 4% body weight) at the end of the first hour of recording. The vertical bars denote the SEM values.

decreased and total sleep time increased over the subsequent 2 h almost to the control levels, but PS remained low throughout the recording period.

(c) Paradoxical sleep deprivation

During the first hour after the 24-h PSD, rats were active and the overall sleep-wakefulness pattern was not significantly different from control. The compensatory increase of PS appeared only in the second hour when the total sleep time reached 73.8% and culminated in the third hour when PS formed 13.9% of the recording time and 24.8% of the total sleep time. Duration of the slow wave sleep almost returned to control level in the fourth hour, but PS remained considerably increased.

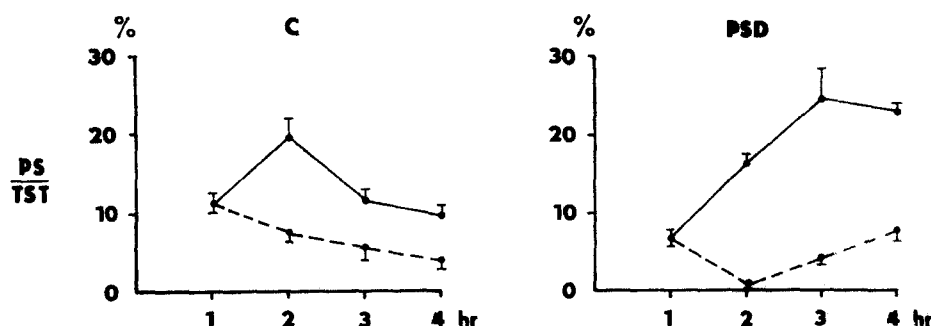


Fig. 4. The effect of paradoxical sleep deprivation and lithium chloride poisoning on the incidence of paradoxical sleep in rats. The ordinate represents the percentage of the total sleep time (TST) covered during the preceding hour by paradoxical sleep. Further description as in Fig. 3.

(d) Paradoxical sleep deprivation and LiCl poisoning

Injection of LiCl after 24-h PSD prevented the development of the compensatory PS increase and elicited marked arousal in the first hour after poisoning (69.6% wakefulness). There was practically no PS during this period. The PS percentage slowly increased over the subsequent 2 h but remained below the control level. The analysis of variance of the mean PS incidence during the h 2–4 showed significant main effects of both poisoning [$F(1, 15) = 73.65$; $P < 0.001$] and PSD [$F(1, 15) = 17.89$; $P < 0.001$] as well as significant poisoning–PSD interaction [$F(1, 15) = 16.95$; $P < 0.01$], but no difference between animals [$F(5, 15) = 0.55$; n.s.].

DISCUSSION

The sleep–wakefulness pattern established in the present experiments under control conditions is slightly different from the values reported in other studies. The slow wave sleep and paradoxical sleep form 50.5% and 5.7% of the total recording time in the untreated rats (a) as compared with 45.0% and 7.0% of Matsumoto et al.⁷, 51.6% and 9.5% of Delacour and Brenot³, 40.4% and 10.8% of Van Twyver¹⁰, 53.1% and 11.8% of Yamazaki et al.¹², 56.2% and 10.8% of Hartmann and Cravens⁵. The difference is mainly due to the relatively short recording time (4-h in the present experiments as compared with 12- and 24-h recording in the above studies) and to incomplete habituation of the animal to the recording situation. As shown by Yamazaki et al.¹², first episodes of PS appear only after 2.4 h of recording in non-habituated rats.

The short habituation may partly account for the observation that the compensatory increase of PS after 24-h PSD becomes apparent only in the second hour of recording. Increased activity of rats immediately after termination of the PSD treatment is a common finding, however¹.

According to Smith et al.⁸ the compensatory increase of PS corresponds to two thirds of the PS debt incurred during PSD. Taking into account that the PS time was approximately doubled in the first 4 h after PSD, the PS debt would be compensated after 16 h at this rate. Since impairment of CTA acquisition induced by 24-h PSD is

fully compensated by 8 h of free sleep between the PSD treatment and CTA learning¹¹, the behavioural compensation is obviously completed at a time when considerable PS debt is still outstanding. This suggests that normal brain function is compatible with substantial depletion of the substances, synthesized during PS. Assuming constant rate of synthesis, concentration of these substances attains about 50% of the normal value after 8 h. According to Danguir and Nicolaidis² CTA acquisition is prevented by 10-h but not 5-h pre-acquisition PSD. If the synthesis of mediators is equal to their uptake under equilibrium conditions, it is not surprising that the duration of PSD leading to the CTA blockade is approximately equal to the minimum duration of free sleep compensating the behavioural consequences of PSD.

The effect of LiCl poisoning on the sleep-wakefulness patterns indicates that PSD does not decrease the effectiveness of the visceral signals of gastrointestinal disturbance caused by this salt which elicits intense arousal lasting for tens of minutes. Slow wave sleep gradually recovers over the next 3 h but PS remains suppressed throughout this interval both in the PS deprived and undeprived rats. These findings seem to be at variance with the statement of Danguir and Nicolaidis² that PS incidence was not affected by LiCl poisoning in their experiments. Although the above authors do not report the electrophysiological results in detail, the difference can be due to experimental variables. The LiCl dosage applied in the present study was much higher than the amount of LiCl spontaneously consumed by the rats of Danguir and Nicolaidis².

The PS reduction induced by LiCl poisoning does not prevent CTA acquisition in undeprived rats but may contribute to the acquisition impairing influence of pre-acquisition PSD. As a consequence of poisoning, the PS debt further increases in the first hours after CS-US association, that is during the period of increased demand for the putative PS-dependent mediators.

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